Fiscal Year:	FY 2020	**Task Last Updated:**	FY 04/28/2020
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	High Dimensional Biology to Understand the Functional Response of Salmonella to Long-Term Multigenerational Growth in the Chronic Stress of Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Tempe	**State:**	AZ
Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	06/01/2015	**End Date:**	12/31/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 12/31/2022 per F. Hernandez/ARC (Ed., 7/27/21) NOTE: Extended to 12/31/2020 per NSSC information (Ed., 7/20/2020) NOTE: Extended to 4/30/2020 per NSSC information (Ed., 6/26/18) NOTE: Extended to 5/31/2019 per F. Hernandez/ARC (Ed., 5/8/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Abu-Ali, Galeb D.V.M., Ph.D. (Harvard College School of Public Health) Barrila, Jennifer Ph.D. (Arizona State University) Huttenhower, Curtis Ph.D. (Harvard College School of Public Health) Ott, Mark Ph.D. (NASA Johnson Space Center) Travisano, Michael Ph.D. (University of Minnesota)		
Grant/Contract No.:	NNX15AL06G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	While unexpected responses in microorganisms are well documented for short duration culture, the impact of long duration, multigenerational growth of microorganisms in the chronic stress of microgravity has not been investigated. The implications of this gap in knowledge are tremendous, as the response to short duration growth in microgravity alters gene expression, pathogenesis-related stress responses (ex., biofilm formation and resistance to antibiotics), and virulence. Moreover, it is known that stress responses (including chronic stress) can increase mutagenesis in bacteria leading to genomic plasticity, which may provide novel mechanistic insight into genomic instability underlying resistance to antibiotics, pathogenicity, and virulence. We propose that long-term growth in spaceflight culture will induce heritable genomic and epigenetic changes in microorganisms with adverse phenotypic outcomes that could have a profound impact on the risk to crew health and vehicle system performance. Our previous studies aboard the Space Shuttle investigated the response of the model enteric pathogen, Salmonella enterica serovar Typhimurium (S. Typhimurium) to short duration spaceflight culture and demonstrated a significant increase in virulence in a murine model of infection, in addition to biofilm formation, as compared to identically grown cultures on the ground. In addition, global microarray and proteomic analyses revealed that 167 transcripts (including small noncoding RNAs/ncRNAs) and 73 proteins changed expression, with the evolutionarily conserved protein Hfq identified as a likely global regulator involved in the response to this environment. Hfq binds to small ncRNAs thereby facilitating their association with mRNAs, the outcome of which plays a diverse role in regulating gene expression, virulence, stress responses, antibiotic resistance, DNA methylation, and genome stability. We hypothesize that long-term multigenerational culture of S. Typhimurium in the spaceflight environment will result in stress-induced mutations and genomic instability as reflected in changes in the genomic, epigenetic, transcriptomic, and select phenotypic characteristics (including virulence) that will reflect a central role for Hfq as a global regulator of these heritable responses. To address this hypothesis, we propose to characterize the impact of long term spaceflight culture (at least 100 generations) of an S. Typhimurium wild type and hfq mutant on the genomic, epigenetic, transcriptomic, virulence, and pathogenesis-related characteristics of these strains. This proposal addresses Research Emphasis 2 - ISS (International Space Station) Microbial Observatory Studies from section B.3 of this announcement, as it is designed to “take advantage of the technological maturity, low cost, and speed of genomic analyses and the rapid generation time of microbes to monitor the evolution of microbial genomic changes in response to the selective pressures present in the spaceflight environment.” In addition, by employing genomic, transcriptomic, and epigenomic analysis, this proposal also addresses the Research Emphasis 1 – Spaceflight Omics Studies in section B.2. In its design, this proposal will address the NASA Space Biology Plan, Overarching Questions CMM-1a, CMM-1d, and CMM5. This proposal also directly addresses the highest priority recommendations from the Decadal Survey for the establishment of an ISS Microbial Observatory conducting long-term, multigenerational studies of microbial population dynamics (Priority P1). In addition, this study analyzes “microbial growth and physiological responses to the multiple stimuli encountered in spaceflight environments,” as described by the panel in Priority P2.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will utilize the microgravity spaceflight platform to 1) to broaden our knowledge of the host-pathogen interaction that leads to infectious disease, and 2) for the development of new therapeutic strategies to combat infectious disease for the general public.
Task Progress:	To accommodate a switch from automated hardware operation to crew manual performance of our experiment on orbit, we successfully completed an SVT (science verification test) to confirm that the Principal Investigator (PI) team can grow and manually transfer Salmonella (wild type/WT and hfq mutant/MT) in the same bags for use in the Exploring Virulence and Omics during Long-term Viability Experiments in Spaceflight (EVOLVES) flight hardware in a fully enclosed container for up to 12 bags (per strain). • Successfully performed experiment verfication test (EVT) EVT-3, which was done to confirm that the PI team can grow and manually transfer Salmonella (WT and MT) in bags housed within sealed Techshot cassettes for up to 24 bags (per strain). • Successful post-EVT optimization to finalize growth times/bag transfer times and nucleic acid yield/integrity for WT and MT. • Excellent RNA and DNA yield and integrity, viable cultures recovered from glycerol – successfully passed all aspects of manual EVT-3. • At NASA’s request to return to automated hardware due to crew time limitations, we performed two engineering tests to optimize updated hardware. Our second test was conducted after COVID-19 limitations on access to laboratory research, which preluded our ability to shake the cultures after bag to bag transfer – however, we successfully met the EVT requirement for the number of bacterial inoculum transfers of the WT strain under static conditions – with excellent turbidity in all bags in one cassette. • Currently preparing for Engineering Test 3 and EVT-3, which will be run simultaneously. • Projected to launch on NG-14 (Northrop Grumman CRS Flight 14), Oct 30, 2020. Invited Presentations: Invited speaker, Nature-NASA Conference on “The Microbiology of Human Spaceflight, ” Johnson Space Center, Houston, TX June 24-27, 2019 Invited speaker, NIH-NASA Summer 2019 Seminar Series, “Microbial pathogen responses to biomechanical forces in infected hosts and microgravity environments,” webinar, July 11, 2019 Invited Keynote Speaker, Commercial Spaceflight Federation annual meeting, Arizona State University, Tempe, AZ, Sept 16, 2019 Invited speaker, ASM Distinguished Lecturer, North Carolina Branch ASM, Greensboro, NC, October 19, 2019

	Invited speaker, “3D Tissues and Microphysiological Systems,” NIH/NCATS pre-workshop, American Society for Gravitational and Space Research Annual Meeting, Denver, CO, Nov 19, 2019 Invited speaker, ASM Distinguished Lecturer, Indiana Branch ASM, Indianapolis, Indiana, April 3-4, 2020 Invited speaker, 21st Century Research – Moving Beyond Animals in the Neurosciences and Infectious Disease Research, Arizona State University, Tempe, AZ, April 24, 2020.
Bibliography Type:	Description: (Last Updated: 07/02/2025)
Books/Book Chapters	Ott CM, Crabbé A, Wilson JW, Barrila J, Castro-Wallace SL, Nickerson CA. "Microbial stress: Spaceflight-induced alterations in microbial virulence and infectious disease risks for the crew." in "Stress Challenges and Immunity in Space: From Mechanisms to Monitoring and Preventive Strategies." Ed. A. Choukèr. Cham: Springer International Publishing, 2020. p. 327-355. https://doi.org/10.1007/978-3-030-16996-1_18 , Jan-2020